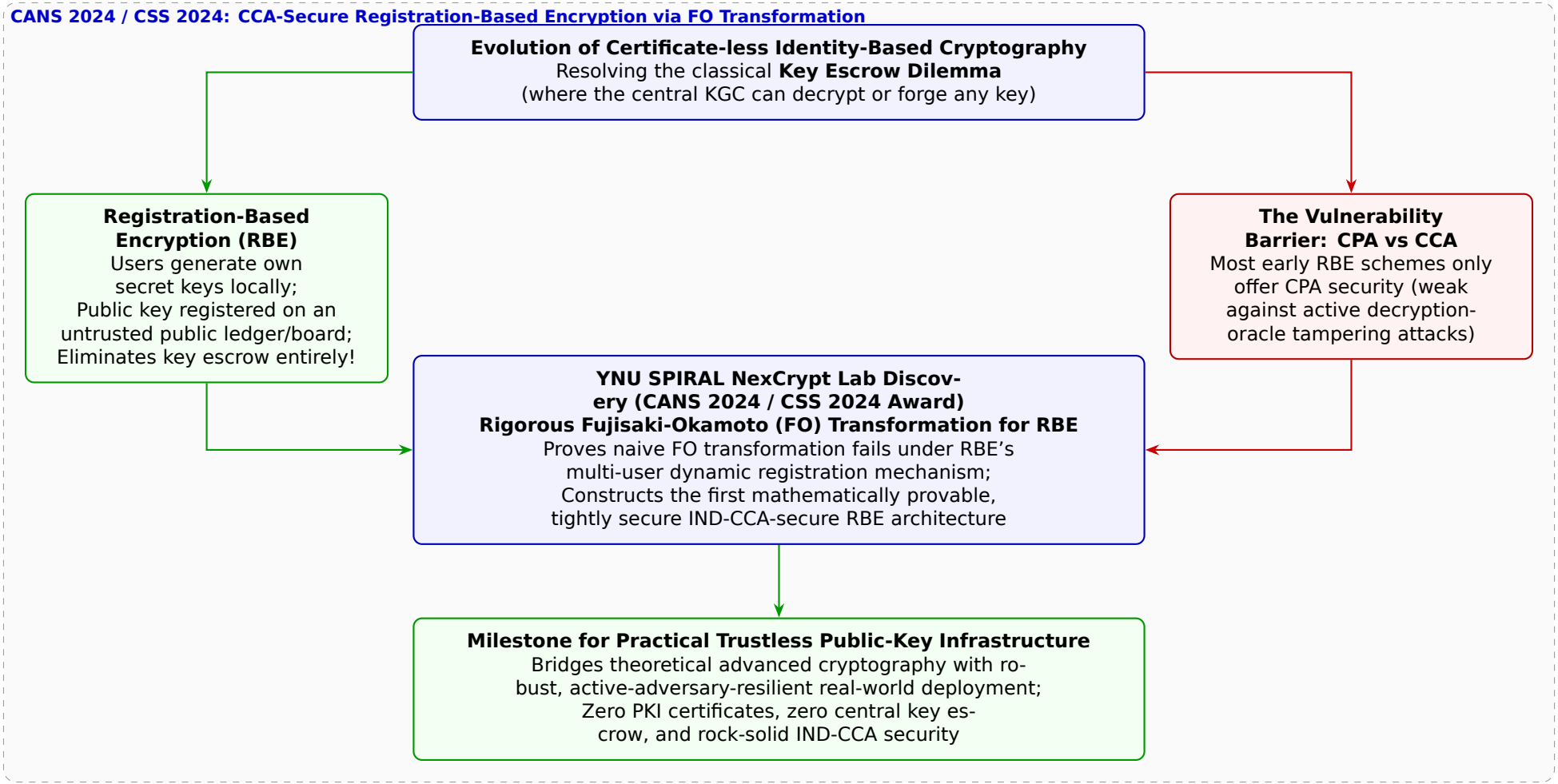
CANS 2024 / CSS 2024: CCA-Secure Registration-Based Encryption via FO Transformation
Evolution of Certificate-less Identity-Based Cryptography
Resolving the classical Key Escrow Dilemma
(where the central KGC can decrypt or forge any key)
Registration-Based Encryption (RBE)
Users generate own secret keys locally;
Public key registered on an untrusted public ledger/board;
Eliminates key escrow entirely!
The Vulnerability Barrier: CPA vs CCA
Most early RBE schemes only offer CPA security (weak against active decryption-oracle tampering attacks)
YNU SPIRAL NexCrypt Lab Discovery (CANS 2024 / CSS 2024 Award)
Rigorous Fujisaki-Okamoto (FO) Transformation for RBE
Proves naive FO transformation fails under RBE's multi-user dynamic registration mechanism;
Constructs the first mathematically provable, tightly secure IND-CCA-secure RBE architecture
Milestone for Practical Trustless Public-Key Infrastructure
Bridges theoretical advanced cryptography with robust, active-adversary-resilient real-world deployment;
Zero PKI certificates, zero central key escrow, and rock-solid IND-CCA security